Friedrich-Alexander-Universität
Technische Fakultät

Pattern Recognition Lab

FAU

Pattern Recognition Lab – A short introduction

模式识别实验室简介

Prof. Dr.-Ing. habil. Andreas Maier, Pattern Recognition Lab

Friedrich-Alexander-Universität Erlangen-Nürnberg, Germany October 2026 ▪ West China Hospital ▪ 华西医院

About the Speaker

Prof. Dr.-Ing. Andreas Maier 主讲人简介

Responsibilities & Experience

- Head of the Pattern Recognition Lab, FAU
- Head of the Competence Center Medical Engineering Erlangen
- Head of the Medical AI Hub of Bavaria
- Member, AI Council of the Free State of Bavaria
- Scientific Counselor, Fraunhofer Institute
- AI Council, German Radiation Protection Society
- European Research Council (grantee & reviewer)
- Visiting Professor at Stanford University and MIT

Industry & Entrepreneurship

- Siemens Healthineers (innovation manager / tech lead)
- Start-ups: *Mira Vision* (shareholder, valued at EUR 10 M); *Norizon* (council, early VC phase)
- Holds more than 15 active US and EU patents

Theory first, application always.

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Profile Center FAU MT

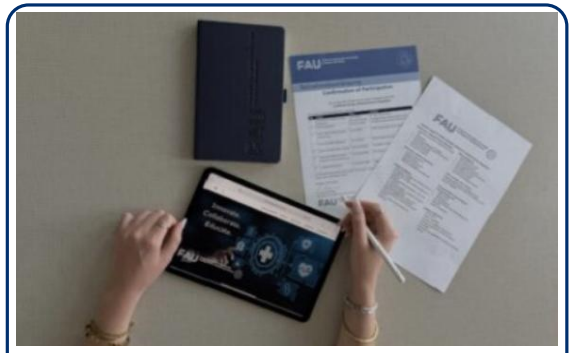
FAU 医学工程重点研究中心

Strategic Goals
战略目标



Research
Shaping the Medicine of Tomorrow

Coordination of interdisciplinary research
Grant consulting



Education
Training Today for the Healthcare of Tomorrow

Development of a medical AI degree program



People
Connecting People, Creating Innovation

Network Activities



Outreach
Uniting Expertise Across Borders

Startup Support

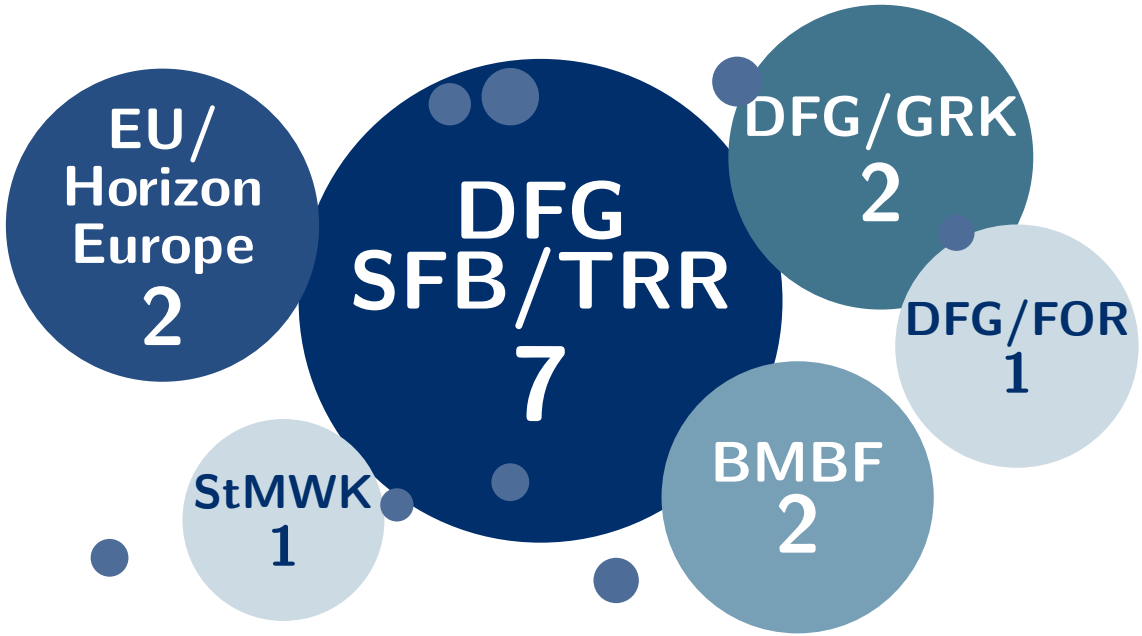
Collaboration Building

Our strategy 合作网络建设：我们的策略

Research Network Analysis Strategies

Visual Mapping of Collaboration Networks

- Map existing collaborations and co-authorships
- Identify central actors



Current large-scale projects of FAU MT members

Building Connections Among Researchers

Organize Thematic Networking Events

- Short pitch sessions to expose researchers to each other's expertise
- Thematic mini-workshops focused on Horizon Europe

Supporting CRC Proposal Development

Transforming Healthcare with Multimodal AI

SFB/CRC 合作研究中心：以多模态人工智能变革医疗

Transforming Healthcare with **Multimodal AI**

Develops **trustworthy, multimodal, and physics-informed AI** that integrates imaging, sensor, omics, and clinical data.

- 30 PIs, from Computer Science, Biology, Clinical, Ethics, among other areas
- 41.4 % women
- 20.7 % junior PIs

Spokespersons



Prof. A. Maier
FAU



Prof. D. Rückert
TUM

FAU + TUM

FAU Friedrich-Alexander-Universität Technische Fakultät

FAU Friedrich-Alexander-Universität Medizinische Fakultät

FAU Friedrich-Alexander-Universität Philosophische Fakultät und Fachbereich Theologie

FAU Friedrich-Alexander-Universität Profilzentrum Medizintechnik

MEDICAL VALLEY · eit Health · HELMHOLTZ MUNICH · DZG DEUTSCHE ZENTREN DER GESUNDHEITSFORSCHUNG · MAX PLANCK GESELLSCHAFT



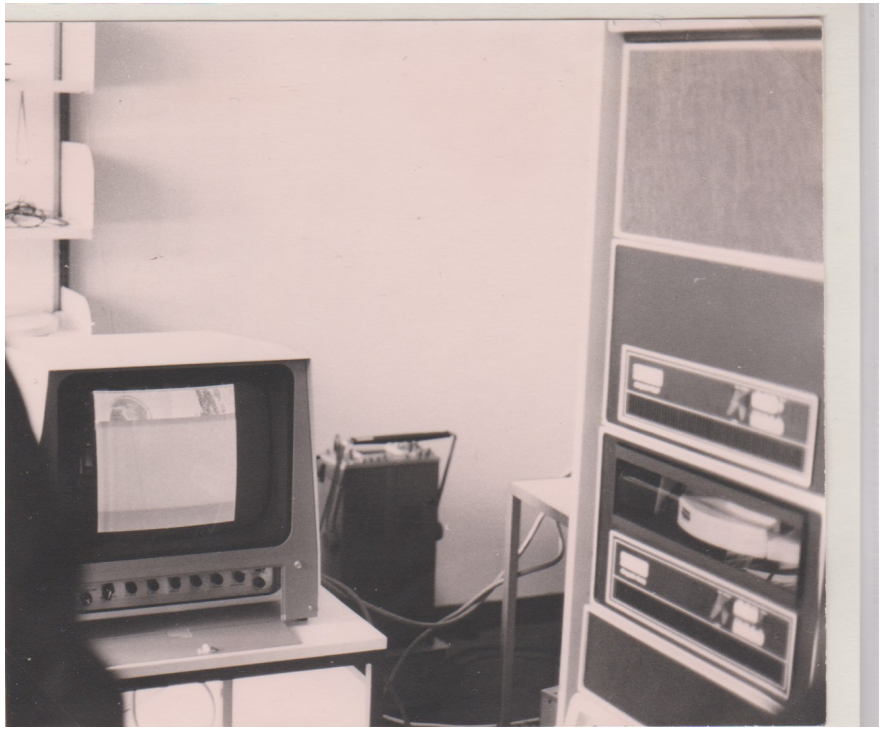
Pattern Recognition Lab

Friedrich-Alexander-Universität Erlangen-Nürnberg 模式识别实验室

Know the Past to Understand the Future

温故而知新

- Founded in 1975, founder: Heinrich Niemann
- Germany's first dedicated Pattern Recognition / Machine Learning Lab
- Medical imaging as early as 1981
- First computer: PDP-11/34 with 248 kB, 2.5 MHz
- Focus: theory & good understanding of hardware



Our Team Today

Pattern Recognition Lab 我们的团队

Secretaries



Alexandra H.



Claudia S.

Head of the Lab



Prof. Dr.-Ing. Andreas Maier

Technician



Sven G.

- ∼60 PhD students & 8 postdocs
- 2 emeritus professors
- ∼3 709 master & bachelor students

- Academic partners in Europe, Asia, North/South America, Australia
- Industry partners: Siemens, Fraunhofer, Adidas, Intel, Audi, IBM, and more

Research Areas

Pattern Recognition Lab 研究方向

Computer Vision

Time Series Intelligence

Magnetic Resonance Imaging

Data Processing for Utility Infrastructure

Learning Algorithms for Medical Big Data Analysis

Speech Processing & Language Understanding

Inverse Problems & Applications

Work at the Lab

实验室的工作





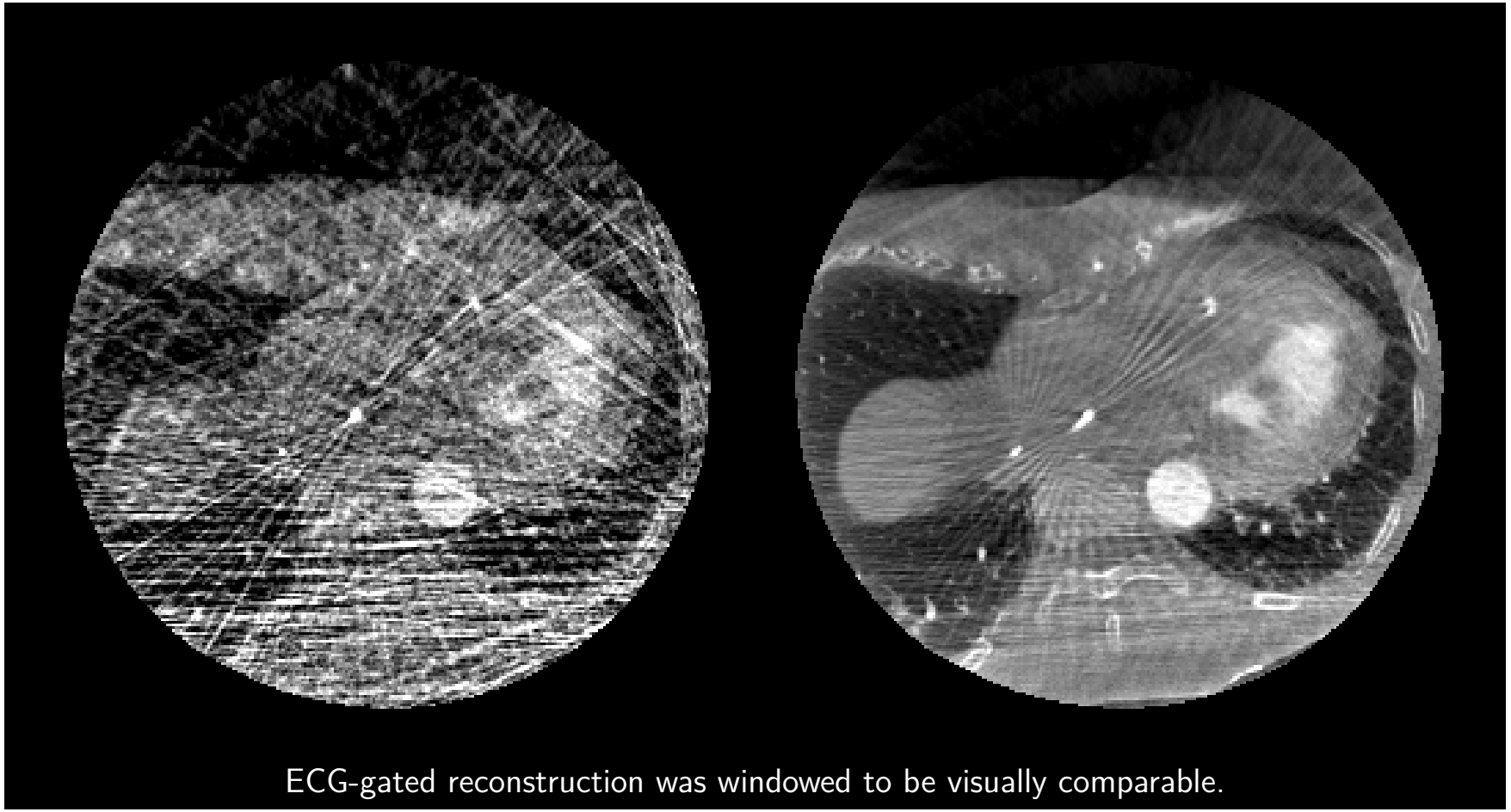






Motion Compensation

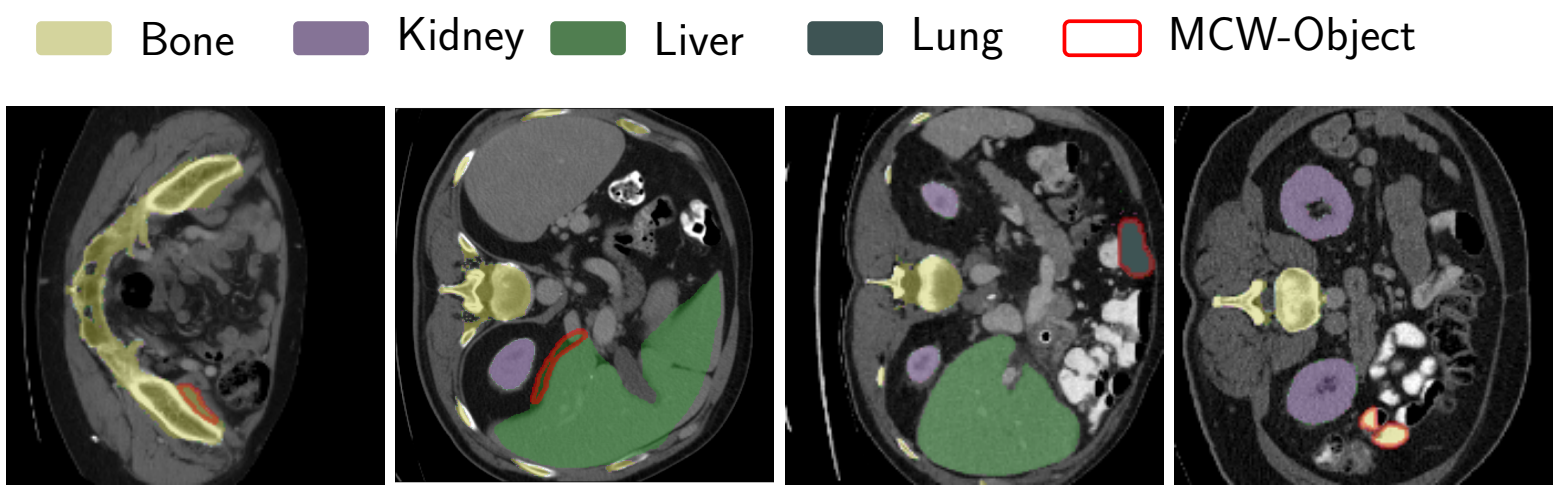
运动补偿

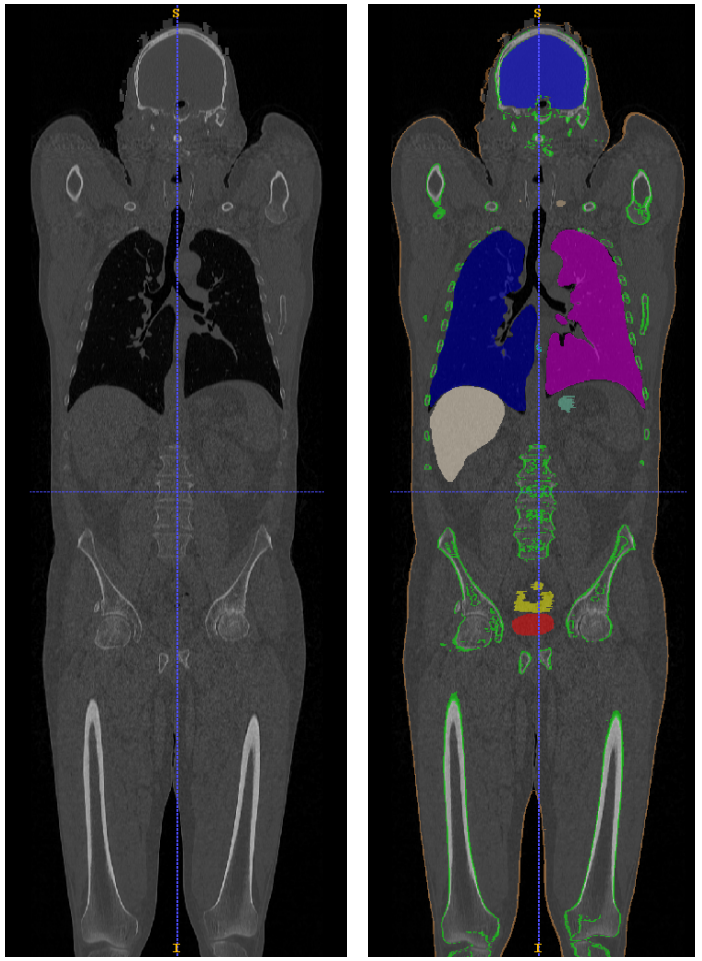


ECG-gated reconstruction was windowed to be visually comparable.

Multi-Organ Segmentation

多器官分割





50 Years in a Nutshell

五十年历程概览

Home > Book

50 Years of Pattern Recognition

The History and Future of the FAU Pattern Recognition Lab

Book | Open Access | Nov 2026

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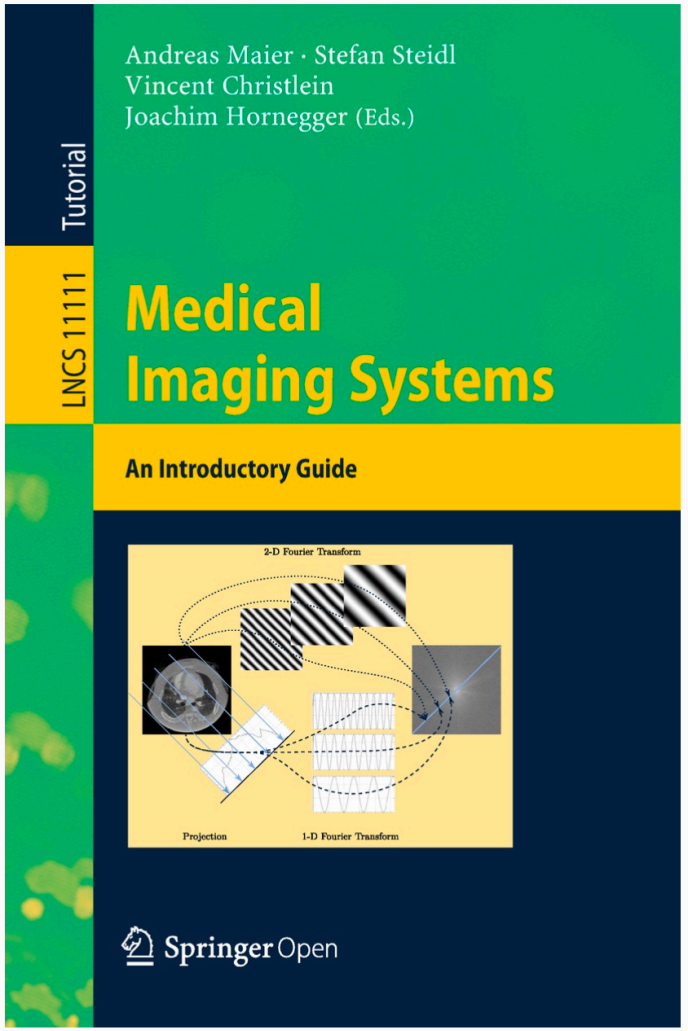
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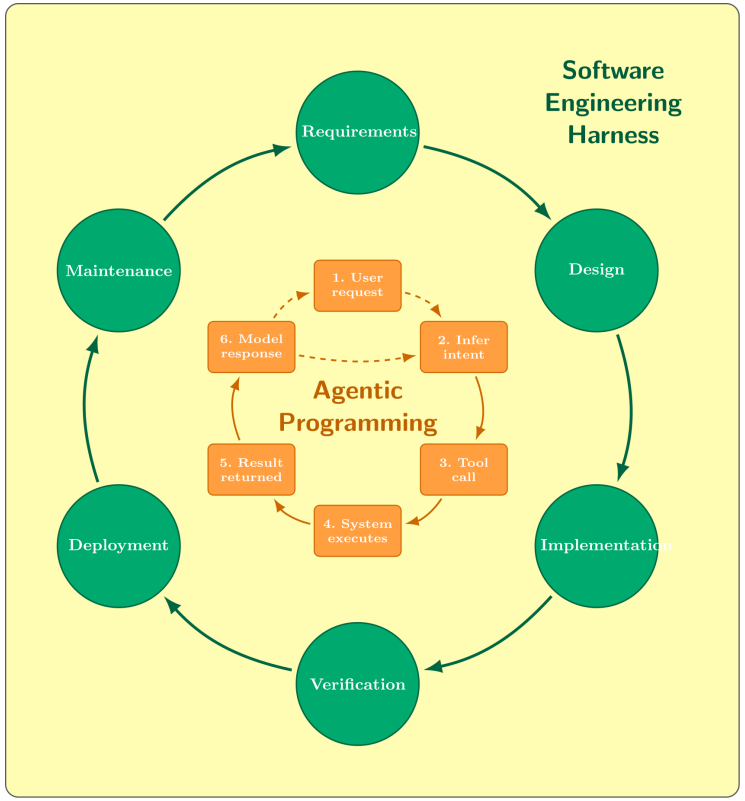
Overview

Editors: Heinrich Niemann, Joachim Hornegger, Andreas Maier, Julian Oelhaf

What is agentic AI?

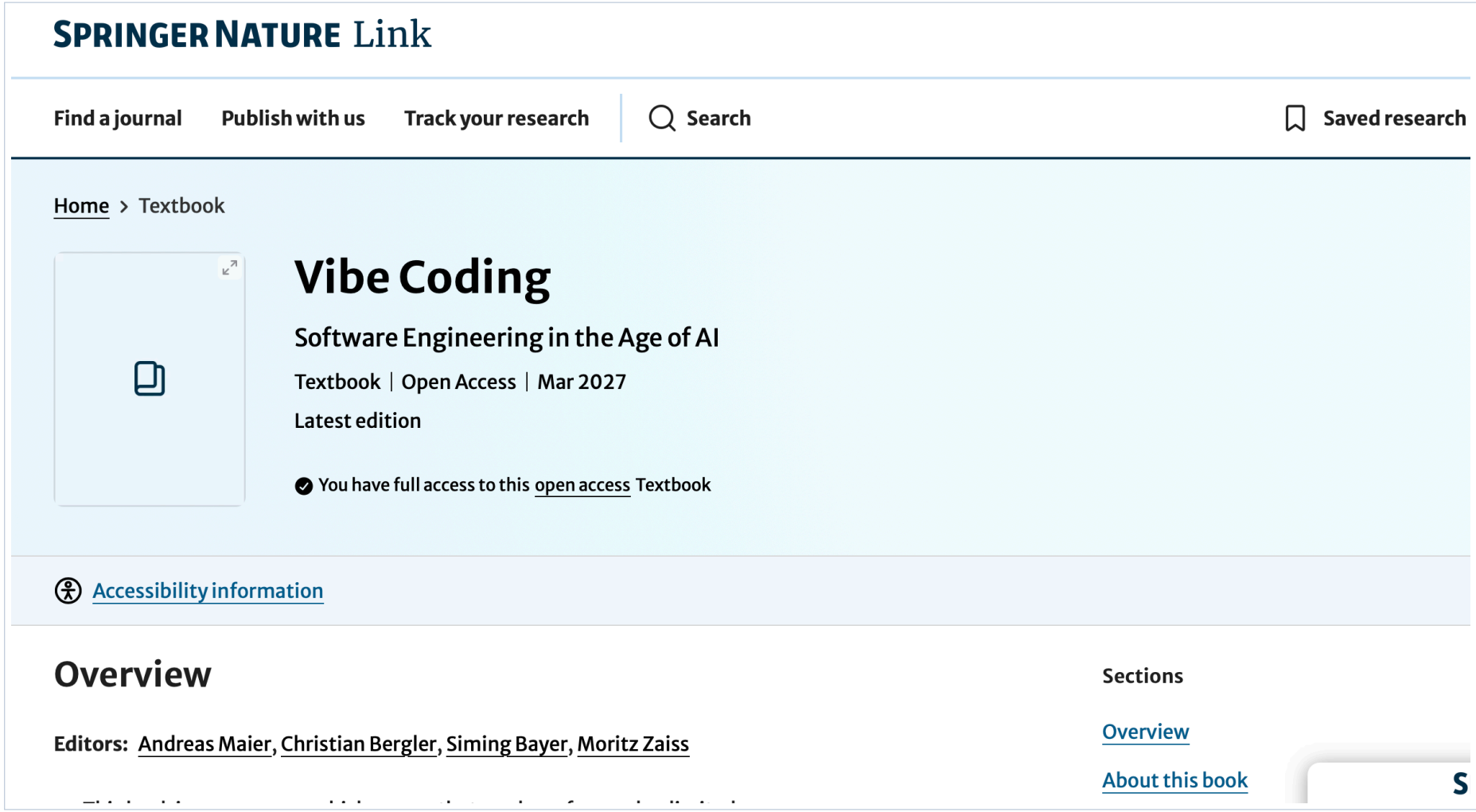
Textbook 什么是智能体 AI?





What is agentic AI?

Textbook 什么是智能体 AI?



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Textbook | Open Access | Mar 2027

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Accessibility information

Overview

Editors: Andreas Maier, Christian Bergler, Siming Bayer, Moritz Zaiss

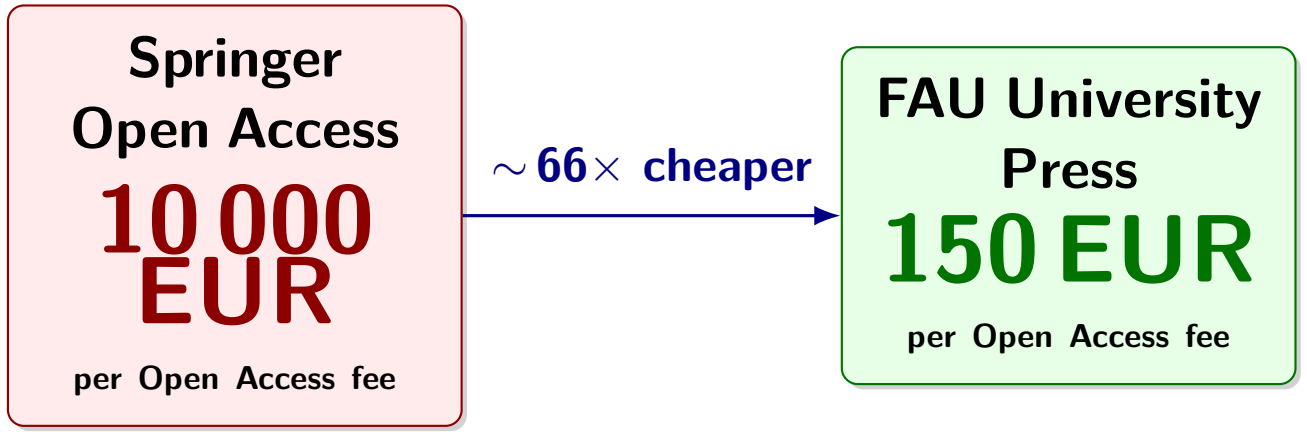
Sections

Overview

About this book

Publishing, Disrupted

Open Access for the price of a pizza party 出版业的变革



- **106** submissions → **93** accepted → one Open Access volume.
- Idea to printed book: **three months**, including **peer review**.

BAIOSPHERE MEDICAL

Speed of Agentic AI

Wall-clock from idea to working artefact　智能体 AI 的速度

3 h Reproduce a paper + code, find new insights, draft a follow-up paper.
arxiv.org/abs/2605.02620

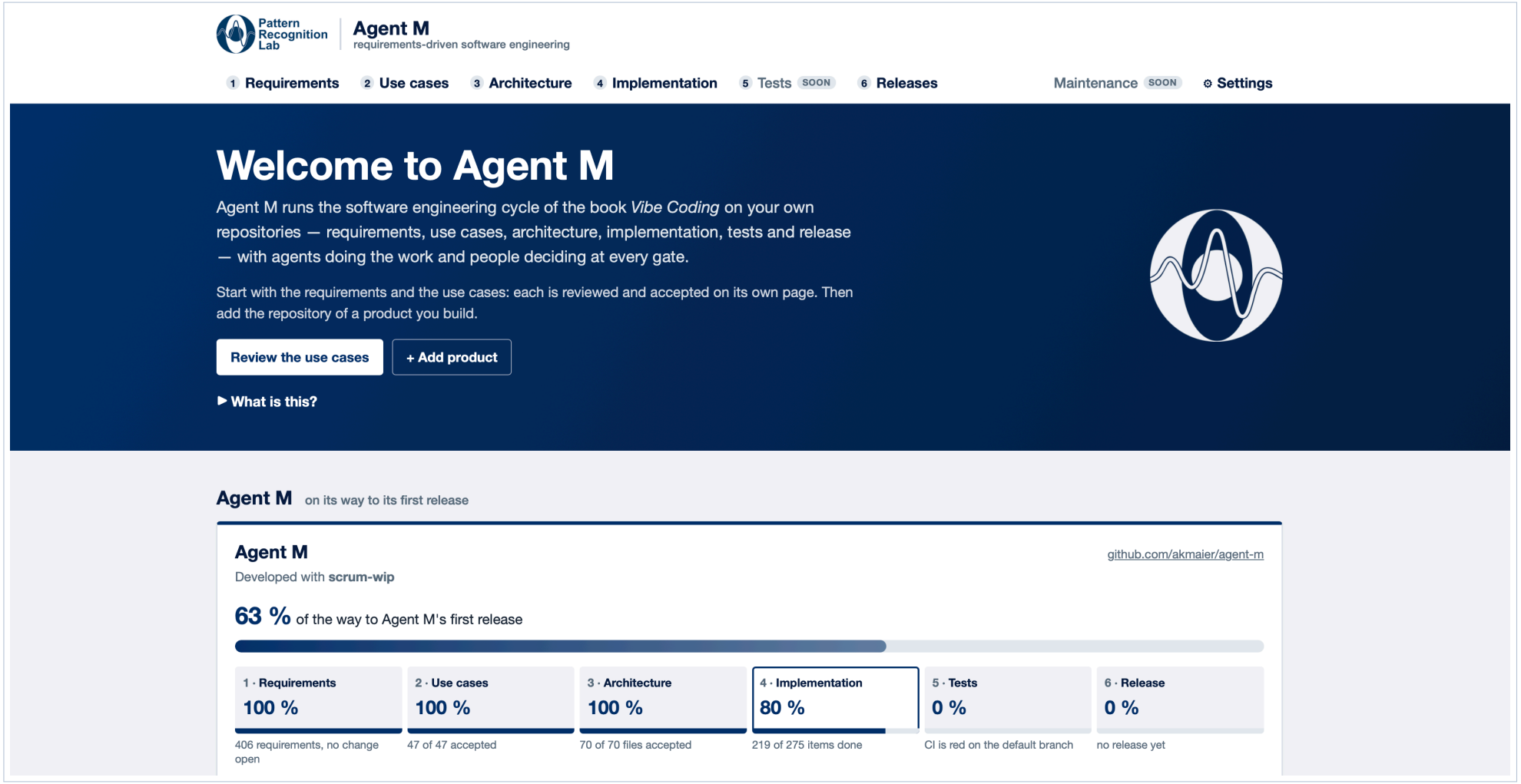
1 d Reproduce a paper *without* its code or data.
arxiv.org/abs/2605.12772

3 d Scrape 200 study programs + ship a RAG chatbot.
github.com/akmaier/IfCampoKnew

1 w Reimplement an entire ATARI VCS emulator – bit-exact, 16 k lines of code.
github.com/akmaier/UnderstandingVCS

Speed of Agentic AI

Wall-clock from idea to working artefact 智能体 AI 的速度



NHR @ FAU – Equipment Snapshot

NHR@FAU 高性能计算中心：设备概览

- Main GPU clusters Alex and Helma:
  - Alex runs A100 40 GB and A40 GPUs.
  - Helma GPUs are Nvidia H100 and H200 accelerators.
- One Helma node is exclusively reserved for LME.
- Total NHR GPU count across projects is 2 000.



Foundation models

Bavarian foundation AI model 基础模型



Cooperations

合作伙伴



Recent new cooperation sites
(in close collaboration with R. Jerecic)

- Shanghai Chest Hospital
- **West China Hospital**
- Peking Union Medical College Hospital
- IIT Bengaluru
- NIMHANS
- Center for Brain Research
- University of Alberta / Edmonton

References

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- Deutscher Zukunftspreis (low-field MRI): https://www.deutscher-zukunftspreis.de
- Conference statistics tooling: https://github.com/akmaier/ConferenceStats
- Agent4CT – agentic re-implementation of deep-learning CT methods: https://github.com/akmaier/Agent4CT
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- **A. Maier**, J. Sopa, G. G. Sahin, P. Perez-Toro and S. Bayer, *Just Ask for a Table: A Thirty-Token User Prompt Defeats Sponsored Recommendations in Twelve LLMs*, arXiv:2605.12772, 2026.
- *IfCampoKnew* – scraper for 200 study programs and an AI RAG chatbot: https://github.com/akmaier/IfCampoKnew
- *UnderstandingVCS* – agent-reimplemented ATARI VCS emulator: https://github.com/akmaier/UnderstandingVCS